Escalation Process Enhancements — Technical Design

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Document type	Technical design & change record
Product / platform	ServiceNow Customer Service Management
Primary table	`sn_customerservice_escalation`
Source instance	`adcomsolutionsdev.service-now.com`
Version	1.0
Status	Deployed on Dev; packaged for migration
Related requirement	SNOW Escalation Widget update v2.1
Consolidated update set	Escalation Enhancements - ALL CHANGES v1.0

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1. Purpose

This document records all configuration and code changes made to the Case Escalation process so the work can be reviewed, migrated, and maintained later.

The enhancements cover three themes:

1. **Request Escalation form fields** — capture richer escalation metadata (level, reason, category, owner, channels, impact, notes).
2. **Customer-visible wording** — rename **Escalation Justification** to **Customer Justification**.
3. **External case notes** — append customer and de-escalation justifications to case **Additional Comments** so customers can see them, without duplicating notes when OOB escalation logic runs.

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2. Scope

In scope

Area	Detail
Table	`sn_customerservice_escalation`
Form view	`request_escalation` (Request Escalation) — opened by **Escalate Case**
Label change	`escalation_justification` → display label **Customer Justification**
Automation	Global business rule posting to case `comments` (Additional Comments)
Migration	Single consolidated update set XML

Out of scope / intentionally unchanged

Item	Reason
Field **#2 Escalation Source**	Already exists as OOB `request_source` (Customer / Internal)
OOB `sn_customerservice.EscalationUtils`	ACL-protected (`sys_policy=read`); not customized
De-escalation form field layout	Existing `request_de_escalation` view retained
SLA “escalation should not satisfy SLA” note from the business doc	Not implemented in this package (configuration/process item outside this change set)
Exclude Escalation from Require Update BR	Separate update set; not in ALL CHANGES v1.0

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3. High-level architecture

```
Case (sn_customerservice_case)
        │
        │  UI Action: Escalate Case
        ▼
Escalation (sn_customerservice_escalation)
  view = request_escalation
  + 8 custom fields (u_*)
```

```
  + Customer Justification (renamed OOB field)
        │
        │  After BR: Append escalation justifications to case
        ▼
Case Additional Comments (comments)
  • "Customer Justification: …"   on escalate / first set
  • "De-escalation Justification: …" on state → Closed Escalated (103)
```

OOB **Escalation field change handler** still writes **internal Work notes** via `EscalationUtils.appendNote()`. This package writes **customer-visible Additional Comments** only.

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4. Dictionary fields added

All custom fields are on **`sn_customerservice_escalation`**, global scope, element names prefixed with `u_`.

#	Label	Element	Type	Choice	Max length	Multi-select
1	Escalation Level / Tier	`u_escalation_level`	Integer	Dropdown (choice=1)	40	No
3	Escalation Reason / Justification	`u_escalation_reason`	String	Dropdown with –None– (choice=3)	255	Yes (`list_collector=true`)
4	Documentation Accuracy Flag	`u_documentation_accuracy`	String	Dropdown (choice=1)	40	No
5	Escalation Category / Type	`u_escalation_category`	String	Dropdown (choice=1)	40	No
6	Owner	`u_escalation_owner`	String	Dropdown (choice=1)	40	No
7	Communication Channel Used for Escalation	`u_communication_channel`	String	Dropdown with –None– (choice=3)	255	Yes (`list_collector=true`)
8	Customer Impact Status	`u_customer_impact_status`	String	Dropdown with –None– (choice=3)	255	Yes (`list_collector=true`)

#	Label	Element	Type	Choice	Max length	Multi-select
9	Additional Notes	`u_additional_notes`	String	None (open text)	1000	N/A

Skipped: #2 Escalation Source — use existing `request_source`.

4.1 Choice values

Escalation Level / Tier (`u_escalation_level`)

Value	Label
0	Level 1
1	Level 2
2	Level 3
3	Level 4

Escalation Reason / Justification (`u_escalation_reason`) — multi-select

Value	Label
0	SLO Breach
1	Repeated Issue
2	Recommended Fix Not Implemented
3	Carrier/Vendor Non-Responsive
4	Technical Complexity
5	Inactivity
6	Business Critical

Documentation Accuracy Flag (`u_documentation_accuracy`)

Value	Label
0	Case Notes Accurate
1	Documentation Error Identified

Escalation Category / Type (`u_escalation_category`)

Value	Label
0	Technical Investigation
1	Performance Optimization
2	Service Restoration
3	Carrier/Vendor Coordination
4	On-Site Dispatch
5	Management Review & Decision
6	Account Management/Billing
7	Architecture/Design Review

Owner (`u_escalation_owner`)

Value	Label
0	NOC
1	Engineering Team
2	Customer Success / Account Management
3	Finance / Billing
4	Executive Management
5	Platform Team

Communication Channel Used for Escalation (`u_communication_channel`) — multi-select

Value	Label
0	Email
1	Phone Call
2	Slack Channel
3	Microsoft Teams
4	Management Review Meeting

Customer Impact Status (`u_customer_impact_status`) — multi-select

Value	Label
0	Severity = Hard Down
1	Intermittent Issues
2	Degraded Performance
3	No Current Impact – Preventive Escalation

Additional Notes (`u_additional_notes`)

Open text, max length **1000** (aligned with OOB `escalation_justification`).

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5. Form layout changes

View

Property	Value
View name	`request_escalation`
View title	Request Escalation
View sys_id	`0359470ab3600300b85d6e5f26a8dc28`
Form section sys_id	`4f590b0ab3600300b85d6e5f26a8dcab`
Table	`sn_customerservice_escalation`

How users reach the form

UI Action **Escalate Case** on `sn_customerservice_case` redirects to:

```
sn_customerservice_escalation.do?
sys_id=-1&sysparm_view=request_escalation&sysparm_view_forced=true&…
```

Field order on Request Escalation (custom fields)

Custom fields were added after the existing OOB fields (for example `request_source`, `reason`, `escalation_justification`).

Position	Element
14	`u_escalation_level`
15	`u_escalation_reason`
16	`u_documentation_accuracy`
17	`u_escalation_category`
18	`u_escalation_owner`
19	`u_communication_channel`
20	`u_customer_impact_status`
21	`u_additional_notes`

Existing OOB elements on the same section (unchanged): `number`, `source_record`, `request_source`, `reason`, `escalation_severity`, `type`, `watch_list`, `add_account_team_members`, `assignment_group`, `assigned_to`, `escalation_justification`, plus form splitters.

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6. Label change — Customer Justification

Property	Before	After
Element	`escalation_justification`	*(unchanged)*
Dictionary column label	Escalation Justification	**Customer Justification**
Field label (`sys_documentation`, en)	Escalation Justification	**Customer Justification**
Plural	Escalation Justifications	**Customer Justifications**
Table	`sn_customerservice_escalation`	*(unchanged)*

Artifacts updated:

- `sys_dictionary` — `sys_dictionary_sn_customerservice_escalation_escalation_justification`
- `sys_documentation` — `sys_documentation_sn_customerservice_escalation_escalation_justification_en`

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7. Business rules

7.1 New — Append escalation justifications to case

Property	Value
Name	Append escalation justifications to case
sys_id	`494966022bbfcb5007a3fa95b891bf10`
Table	`sn_customerservice_escalation`
When	after
Order	200
Insert	true
Update	true
Active	true
Scope	global
Update name	`sys_script_494966022bbfcb5007a3fa95b891bf10`
Repo artifact	`servicenow/business-rules/append-escalation-justifications-to-case-comments.js`

Behavior

Event	Condition	Writes to case
Escalate	Customer Justification first set (insert, or empty → value)	Additional Comments: `Customer Justification: {text}`
De-escalate	State changes to **103** (Closed Escalated) and de-escalation justification is populated	Additional Comments: `De-escalation Justification: {text}`

Only source tables **`sn_customerservice_case`** and **`csm_order_case`** are updated. Notes go to **`comments`** (customer-visible), not `work_notes`.

Duplicate-note guard

OOB **Escalation field change handler** (order **1000**) calls `EscalationUtils.appendNote()`, which performs a nested `escalation.update()` for internal work notes. That nested update re-fired this business rule and posted Customer Justification twice.

Fix: before posting, query `sys_journal_field` for the same case `comments` entry containing the marker text; skip if already present.

7.2 OOB rules that interact (not modified)

Name	Role in the flow
Escalation field change handler	After insert/update; calls `EscalationUtils.appendNote` / `notifyUsers` for state, severity, trend, work notes, comments
Append escalation approval comments	Before; copies approval comments into escalation work notes
Close escalation	Before; sets `de_escalated_by`
Escalation state management	Before; case/escalation state coordination

`EscalationUtils` could not be customized (ACL: *ACL Exception Update Failed due to security constraints*). Customer-visible note behavior is therefore implemented in the global BR above.

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8. Update sets / migration

Recommended: one consolidated set

Property	Value
Name	Escalation Enhancements - ALL CHANGES v1.0
File	`updatesets/Escalation_Enhancements_ALL_CHANGES_v1.0.xml`
Change count	27
Contents	8 field dictionaries + labels + choices, form layout, Customer Justification rename, final business rule (deduped)

Target import steps

1. **Retrieved Update Sets → Import Update Set from XML**
2. Select `Escalation_Enhancements_ALL_CHANGES_v1.0.xml`
3. **Preview**
4. Resolve any collisions if prompted
5. **Commit**

Prerequisite on target: Customer Service Management / Case Escalation (`sn_customerservice_escalation`) available.

Historical split sets (only if not using ALL CHANGES)

Apply in this order:

Order	Update set	File	Changes
1	Escalation Form Fields v1.0	*(on Dev; captured in ALL CHANGES)*	24
2	Escalation Customer Notes v1.0	`updatesets/Escalation_Customer_Notes_v1.0.xml`	3
3	Escalation Customer Notes Dedup v1.1	`updatesets/Escalation_Customer_Notes_Dedup_v1.1.xml`	1

Prefer **ALL CHANGES v1.0** so this sequence is unnecessary.

Not included in ALL CHANGES

Update set	Notes
Exclude Escalation from Require Update BR	Separate; rebuild with `npm run consolidate:escalation -- --include-require-update` if needed

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9. Repository deliverables

Path	Purpose
`scripts/deploy-escalation-fields.js`	Create/idempotent fields + choices on Dev

Path	Purpose
`scripts/deploy-escalation-form.js`	Add fields to `request_escalation` form
`scripts/deploy-escalation-customer-notes.js`	Label rename + BR deploy + XML export
`scripts/consolidate-escalation-update-set.js`	Build ALL CHANGES XML
`servicenow/business-rules/append-escalation-justifications-to-case-comments.js`	BR source of truth
`updatesets/Escalation_Enhancements_ALL_CHANGES_v1.0.xml`	**Migration package**
`updatesets/Escalation_Customer_Notes_v1.0.xml`	Historical split set
`updatesets/Escalation_Customer_Notes_Dedup_v1.1.xml`	Historical split set

npm scripts

```
npm run deploy:escalation-fields
npm run deploy:escalation-form
npm run deploy:escalation-customer-notes
npm run consolidate:escalation
```

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10. Test evidence (Dev)

Case	Escalation	Result
CS1703276	ESC0004503 / ESC0004504	Early tests; showed duplicate Customer Justification before dedup fix
CS1703308	ESC0004505	After dedup: **1** Customer Justification + **1** De-escalation Justification on Additional Comments

How to verify on a case

1. Open the case form.
2. Review **Additional Comments** (external / customer-visible) — not **Work notes**.

3. Expect:
    - After escalate: `Customer Justification: …`
    - After de-escalate (state 103): `De-escalation Justification: …`
4. Confirm Customer Justification appears **once**.

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11. Operational notes

- Multi-select fields use ServiceNow choice `list_collector` attribute; UI renders a multi-select control on the form.
- Creating escalations via REST still fires the after BR; UI Escalate Case is the primary intended path.
- Internal work-note messages from OOB (`Escalation … has been approved/closed`) remain unchanged.
- If Customer Justification is filled later on an existing escalation (empty → value), the BR posts once at that time.

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12. Change log

Version	Date	Summary
1.0	2026-09-11	Eight form fields + Request Escalation form layout
1.0	2026-10-09	Customer Justification rename + external notes BR
1.1	2026-10-13	Dedup guard for duplicate Customer Justification notes
1.0 ALL	2026-10-14	Consolidated update set for migration
Doc 1.0	2026-10-14	This technical design document

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13. Appendix — Requirement mapping (SNOW Escalation Widget update v2.1)

Req item	Requirement	Implementation
1		`u_escalation_level`

Req item	Requirement	Implementation
	Escalation Level / Tier dropdown	
2	Escalation Source	Skipped — OOB `request_source`
3	Escalation Reason / Justification multi-select	`u_escalation_reason`
4	Documentation Accuracy Flag	`u_documentation_accuracy`
5	Escalation Category / Type	`u_escalation_category`
6	Owner	`u_escalation_owner`
7	Communication Channel multi-select	`u_communication_channel`
8	Customer Impact Status multi-select	`u_customer_impact_status`
9	Additional Notes open text	`u_additional_notes`
—	Rename Escalation Justification → Customer Justification	Dictionary + documentation
—	Append escalation notes to external case notes	BR → case `comments`
—	Append de-escalation notes to external case notes	BR → case `comments` on state 103

End of document.